Replicating Stambaugh (1999), *Predictive Regressions*

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Abstract

Stambaugh (1999) shows that the ordinary least squares slope in a return predictability regression is biased upward in finite samples when the predictor is persistent and its innovations are negatively correlated with return innovations—the defining features of the dividend–price ratio. We rebuild the data from CRSP, replicate the paper's Table 1, Table 2, and Figure 1 within a stated tolerance, and extend every exhibit through 2024. Our persistence estimates match the paper to the third decimal, our finite-sample p-value for 1927–1996 is 0.177 against the paper's 0.17, and all sixteen Bayesian posterior cells match within roughly 0.03. Extending the sample, we find the problem the paper identifies has grown more severe: in 1997–2024 the naive p-value is 0.015 while the finite-sample p-value is 0.149, a tenfold gap.

1 The problem

The standard test of return predictability regresses next period's excess return on this period's dividend–price ratio,

$$r_{t+1} = \alpha + \beta x_t + u_{t+1}, \qquad x_{t+1} = \theta + \rho x_t + v_{t+1}, \tag{1}$$

where x_t is the dividend–price ratio and the innovations (u_{t+1}, v_{t+1}) are jointly distributed with covariance matrix Σ. OLS is consistent here, but it is not unbiased: strict exogeneity fails, because the return innovation u_{t+1} is correlated with the yield innovation v_{t+1}, which in turn moves the *future* regressor.

The bias arrives in two steps. First, the least squares estimator of an AR(1) coefficient is biased downward by approximately $(1+3\rho)/T$ (Kendall, 1954); with ρ near one this is substantial. Second, that error transmits into the predictive slope with its sign reversed,

$$E[\hat{\beta} - \beta] = \frac{\sigma_{uv}}{\sigma_{vv}}\, E[\hat{\rho} - \rho] \approx -\frac{\sigma_{uv}}{\sigma_{vv}} \cdot \frac{1+3\rho}{T}. \tag{2}$$

Because a price increase raises the return and lowers the yield in the same month, $\sigma_{uv} < 0$, and the slope bias is therefore *positive*. Both ingredients are necessary: remove the persistence or the innovation correlation and the bias disappears. In our data both are present in abundance—persistence of 0.99 and an innovation correlation of -0.95.

2 Data

Two WRDS pulls supply everything.

Table 1: Summary statistics for the monthly excess return, the dividend-price ratio in levels, and the log dividend-price ratio. The AR(1) column shows that log D/P is highly persistent, which is central to the Stambaugh finite-sample bias.

Sample	Start	End	Obs.	Mean excess return (%)	Std. excess return (%)	Mean D/P	Std. D
Full sample	1927-06	2024-12	1171	0.515000	5.342000	0.035000	0.016
1927-1996	1927-06	1996-12	835	0.499000	5.590000	0.042000	0.014
1927-1951	1927-06	1951-12	295	0.494000	7.551000	0.053000	0.016
1952-1996	1952-01	1996-12	540	0.502000	4.153000	0.036000	0.009
1977-1996	1977-01	1996-12	240	0.538000	4.314000	0.036000	0.009

Market index. We use `crsp.msi`, CRSP's pre-aggregated monthly value-weighted market index, rather than the security-level stock file. CRSP has already performed the capitalization weighting, so we consume a finished series instead of reconstructing it from thousands of monthly security-level observations, which would require lagged market-capitalization weights and delisting-return handling.

Dividends without a dividend database. The index file reports the value-weighted return both including dividends (`vwretd`) and excluding them (`vwretx`). Their difference is the month's dividend as a fraction of the prior month's price. Compounding the price-only return yields a price level P_t, so the monthly dividend in index units is $D_t = (\text{vwretd}_t - \text{vwretx}_t)P_{t-1}$; summing twelve trailing months gives the annual dividend, and the predictor is $D_{12,t}/P_t$.

Risk-free rate. From `ff.factors_monthly`. The dependent variable is the continuously compounded excess return, $\log(1 + R_m) - \log(1 + R_f)$, matching the paper's Table 1 specification.

Sample. The merged panel spans June 1927 to December 2024 (1,171 monthly observations). It begins in mid-1927 because the risk-free series starts in July 1926 and the trailing-twelve-month dividend window consumes a further eleven months, leaving our replication samples about six observations shorter than the paper's.

Index universe. Stambaugh uses a NYSE-only value-weighted index. Our WRDS instance provides no pre-built NYSE-only monthly index carrying both `vwretd` and `vwretx`, so we use the CRSP total-market value-weighted index. Because both are value-weighted, the same large-capitalization firms—predominantly NYSE-listed—dominate each, and for the first half of the sample the universes effectively coincide, since AMEX enters CRSP in 1962 and NASDAQ at the end of 1972. As a robustness check we rebuilt a NYSE-only value-weighted index directly from the CRSP monthly stock file, weighting each security's return by its lagged market capitalization. Over the full overlapping sample the two dividend-price series correlate at 0.994, and the predictive slopes differ by 0.014 (0.129 for the total-market index against 0.143 for the NYSE-only index); Table 7 reports the comparison. The index choice does not materially affect the results.

Table 1 and Figure 1 describe the resulting panel.

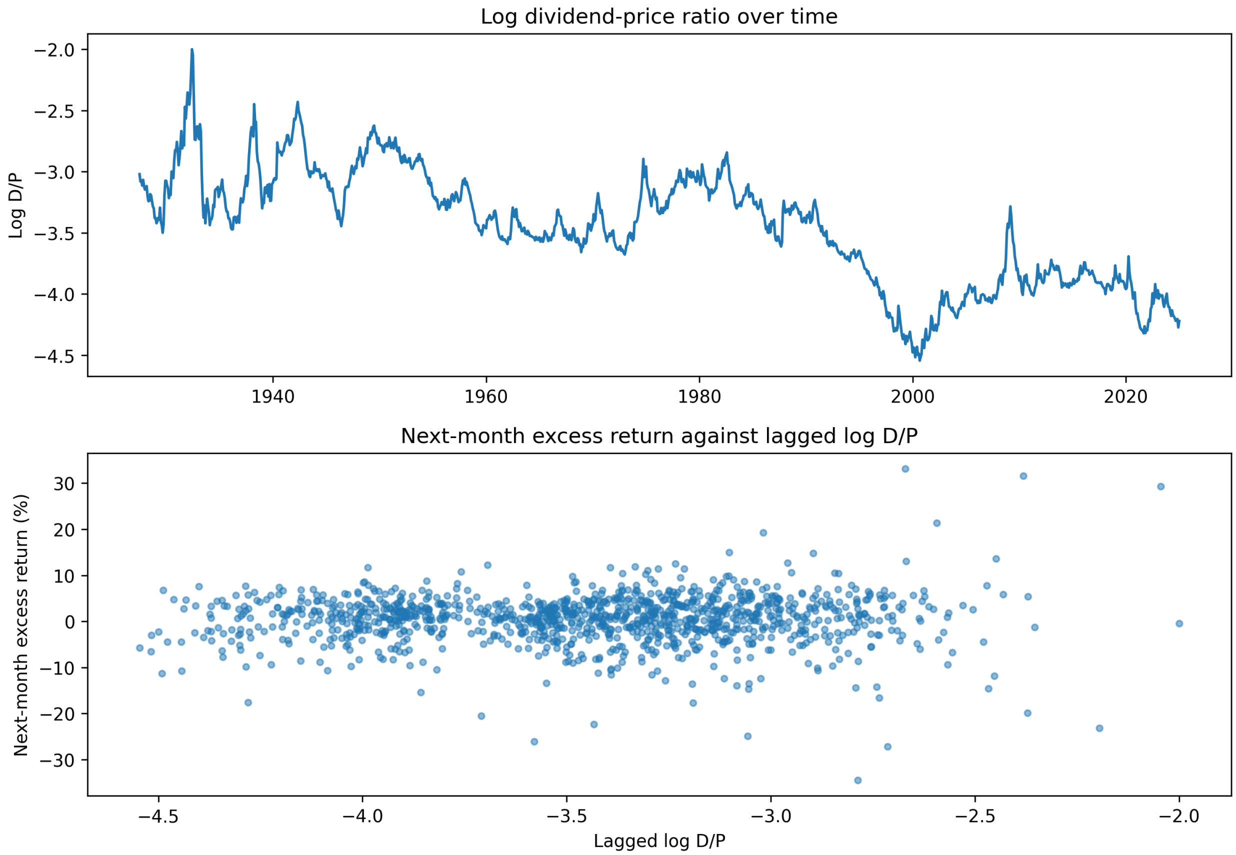

Figure 1: The log dividend-price ratio is highly persistent, while the scatter of next-month excess returns against lagged log D/P is noisy. This is the core empirical setting in which Stambaugh's finite-sample bias matters: the predictor moves slowly, but the return relationship is difficult to see cleanly in monthly data.

Table 2: Finite-sample properties of the OLS predictive slope (replication of Stambaugh 1999, Table 1). Part A reports the simulated distribution of the OLS slope when the true slope is zero, using the parameters of Part C (20,000 replications): the estimator is biased upward, right-skewed, and fat-tailed. Part B reports the same properties as asserted by the standard regression model – unbiased, symmetric, normal – with the textbook standard error and naive p-value. Part C reports the sample characteristics that drive the gap: a predictor with persistence near one whose innovations are strongly negatively correlated with return innovations ($\sigma_{uv} < 0$). Takeaway: the honest p-values in Part A are roughly three to ten times the naive ones in Part B – most starkly in 1952–1996, where the naive test rejects no-predictability at 1.4% while the finite-sample test cannot reject it at 16.5%.

		1927-1996	1927-1951	1952-1996	1977-1996
A. Simulated (true $\beta = 0$)	Bias	0.07	0.19	0.20	0.52
	Standard deviation	0.16	0.34	0.29	0.51
	Skewness	0.74	0.84	1.04	1.33
	Kurtosis	3.90	4.12	4.83	5.89
	p-value for $\beta = 0$	0.18	0.41	0.17	0.72
B. Standard regression	Bias	0.00	0.00	0.00	0.00
	Standard deviation	0.14	0.28	0.21	0.31
	Skewness	0.00	0.00	0.00	0.00
	Kurtosis	3.00	3.00	3.00	3.00
	p-value for $\beta = 0$	0.06	0.21	0.01	0.27
C. Sample characteristics	$\hat{\beta}$	0.21	0.22	0.46	0.19
	T	834	294	539	239
	ρ	0.973	0.946	0.981	0.990
	$\sigma_u^2 \times 10^4$	31.22	57.26	17.15	18.65
	$\sigma_v^2 \times 10^4$	0.112	0.265	0.027	0.030
	$\sigma_{uv} \times 10^4$	-1.672	-3.564	-0.647	-0.704

3 Replication results

Table 2 reports the finite-sample properties of the OLS predictive slope for each of the paper's four subsamples, in the paper's own three-part structure.

Part C of Table 2 reproduces the paper's parameter estimates closely: our persistence estimates of 0.973, 0.946, 0.981, and 0.990 compare to the paper's 0.972, 0.948, 0.980, and 0.987, and the innovation covariances match in sign, magnitude, and cross-sample pattern. Part A, simulated at those parameters under the null $\beta = 0$, reproduces the paper's finite-sample bias (0.074 against 0.07 in the full sample), standard deviation (0.155 against 0.16), skewness (0.74 against 0.71), kurtosis (3.90 against 3.84), and finite-sample p-value (0.177 against 0.17). The contrast between Parts A and B is the paper's central point: the standard regression model asserts an unbiased, symmetric, normal estimator and delivers a p-value of 0.06, while the actual finite-sample distribution is shifted, right-skewed, fat-tailed, and yields 0.18.

Table 3 reports Bayesian posterior moments under the paper's four specifications. Parts A and B use the likelihood conditioned on x_0; parts C and D use the exact likelihood, which treats x_0 as a draw from the predictor's stationary distribution, $x_0 \sim N\big(\theta/(1-\rho), \sigma_v^2/(1-\rho^2)\big)$.

Under a flat prior with the conditional likelihood (Part A) the posterior is conjugate and centers on the OLS estimate; the posterior mean of 0.21 and $P(\beta \leq 0) = 0.06$ match the paper exactly and

Table 3: Posterior distributions for β (replication of Stambaugh 1999, Table 2). Parts A and B condition the likelihood on x_0; with a flat prior the posterior is conjugate and centers on the OLS estimate, and Part B adds the stationarity restriction $\rho \in (-1, 1)$, which binds only in the short, highly persistent 1977–1996 sample. Parts C and D use the exact likelihood, treating x_0 as a draw from the predictor's stationary distribution; that term is nonlinear in ρ, so these posteriors are not conjugate and are sampled by Metropolis-Hastings. Part D applies the paper's alternative prior. Takeaway: the exact likelihood strengthens the evidence for predictability while Part D's prior pulls back toward ρ near one, and across all four specifications the posterior places far less mass below zero than the finite-sample frequentist p-values of Table 2 would suggest.

		1927-1996	1927-1951	1952-1996	1977-1996
A. Conditional; ρ free	Mean	0.21	0.22	0.46	0.18
	Std. Dev.	0.14	0.28	0.21	0.30
	Skewness	0.02	0.02	0.02	0.03
	Kurtosis	3.02	3.02	3.00	3.03
	Prob($\beta \leq 0$)	0.06	0.21	0.01	0.28
B. Conditional; $\lvert\rho\rvert < 1$	Mean	0.21	0.22	0.46	0.28
	Std. Dev.	0.14	0.28	0.20	0.24
	Skewness	0.03	0.06	0.14	0.54
	Kurtosis	2.99	2.95	2.86	3.17
	Prob($\beta \leq 0$)	0.06	0.21	0.01	0.11
C. Exact; $\lvert\rho\rvert < 1$	Mean	0.23	0.24	0.34	0.34
	Std. Dev.	0.14	0.26	0.16	0.20
	Skewness	0.11	0.14	0.43	0.12
	Kurtosis	2.87	2.76	2.83	2.75
	Prob($\beta \leq 0$)	0.04	0.18	0.00	0.04
D. Exact, alt. prior	Mean	0.17	0.16	0.29	0.25
	Std. Dev.	0.14	0.27	0.16	0.25
	Skewness	-0.20	-0.03	-0.09	0.55
	Kurtosis	2.82	3.12	2.34	2.72
	Prob($\beta \leq 0$)	0.13	0.28	0.03	0.15

provide an internal check that the sampler is correct. Part B imposes stationarity, $\rho \in (-1, 1)$, by rejection. The restriction is nearly non-binding in the long samples but discards roughly one draw in five in 1977–1996, where it raises the posterior mean and halves $P(\beta \leq 0)$ from 0.28 to 0.11 (the paper reports 0.26 to 0.13). The prior matters precisely where the data cannot identify persistence on their own.

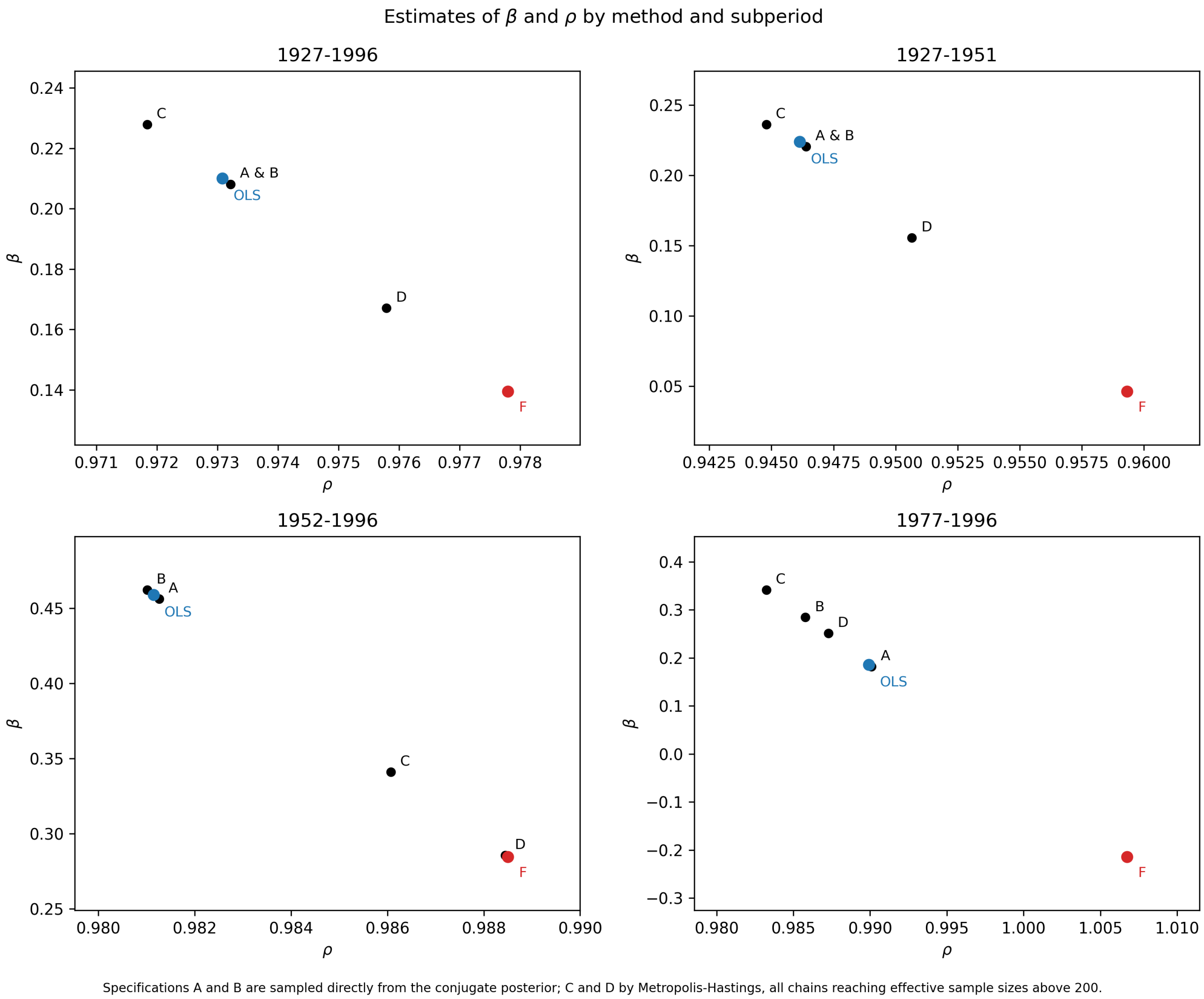

Figure 2: Estimates of β and ρ by subsample and method. In every panel the bias-adjusted estimate lies below and to the right of the OLS estimate: correcting the finite-sample bias simultaneously lowers the slope and raises the persistence, because both are manifestations of the same problem. The 1977–1996 panel reproduces the paper's most extreme result—the bias-corrected persistence exceeds unity and the adjusted slope turns negative—which is the correction reporting that this short, near-unit-root sample cannot identify the parameters, and which is precisely the region the Bayesian stationarity restriction excludes by construction.

The exact likelihood in Parts C and D is nonlinear in ρ and therefore not conjugate, so these two specifications are sampled by random-walk Metropolis-Hastings. The chain moves in an unconstrained parameterization—the regression coefficients together with $\log \sigma_u$, $\log \sigma_v$, and $\operatorname{atanh} \operatorname{corr}(u, v)$—so every proposal yields a valid covariance matrix, with the corresponding Jacobian included in the log target. The proposal covariance is estimated from a short pilot chain and

Table 4: Finite-sample properties of the OLS predictive slope, samples extended through 2024. Structure follows Table 2: Part A simulates the null at the Part C parameters, Part B gives the standard regression model's assertions. The 1997–2024 column lies entirely outside Stambaugh's original sample. Takeaway: the bias falls as the sample lengthens (0.07 to 0.06 in the full sample, tracking the $1/T$ rate), but in the modern subsamples it grows large relative to the slope itself, and the gap between the naive and finite-sample p-values widens to tenfold in 1997–2024 (0.01 against 0.15).

		1927-2024	1952-2024	1997-2024	2005-2024
A. Simulated (true $\beta = 0$)	Bias	0.06	0.14	0.57	0.74
	Standard deviation	0.11	0.20	0.86	1.08
	Skewness	0.79	1.03	0.99	0.96
	Kurtosis	4.11	4.64	4.57	4.37
	p-value for $\beta = 0$	0.24	0.30	0.15	0.31
B. Standard regression	Bias	0.00	0.00	0.00	0.00
	Standard deviation	0.10	0.13	0.65	0.85
	Skewness	0.00	0.00	0.00	0.00
	Kurtosis	3.00	3.00	3.00	3.00
	p-value for $\beta = 0$	0.09	0.05	0.01	0.09
C. Sample characteristics	$\hat{\beta}$	0.13	0.21	1.43	1.13
	T	1170	875	335	239
	ρ	0.984	0.990	0.969	0.956
	$\sigma_u^2 \times 10^4$	28.54	18.98	21.64	20.32
	$\sigma_v^2 \times 10^4$	0.083	0.021	0.010	0.011
	$\sigma_{uv} \times 10^4$	-1.317	-0.564	-0.424	-0.455

scaled by the usual $2.4/\sqrt{d}$ rule, because β and ρ are strongly correlated in this posterior and a diagonal random walk mixes poorly along that ridge. We validate the sampler by running it on the *conditional* likelihood with a flat prior, which is exactly the conjugate setting of Part A: it recovers a posterior mean of 0.209, standard deviation 0.135, and $P(\beta \leq 0) = 0.059$ against the conjugate values 0.21, 0.14, and 0.06.

The two adjustments push in opposite directions. The exact likelihood is informative about ρ and strengthens the evidence for predictability: for 1927–1996 the posterior mean rises to 0.23 and $P(\beta \leq 0)$ falls to 0.04, against the paper's 0.23 and 0.05. Part D's alternative prior, $p(b, \Sigma) \propto (1-\rho^2)^{-1}\sigma_v^2|\Sigma|^{-5/2}$, places more weight near $\rho = 1$ and pulls back: the mean falls to 0.17 and $P(\beta \leq 0)$ rises to 0.13, against the paper's 0.19 and 0.10. All sixteen posterior cells agree with the paper to within about 0.03. Chains were run to effective sample sizes above 200, which supports the means and tail probabilities to the two decimals reported; the higher moments should be read as indicative only.

Figure 2 reproduces the geometry of the paper's Figure 1. Our bias-adjusted points correspond to the paper's "F" points and match closely: $(\rho, \beta) = (0.978, 0.14)$ against the paper's $(0.976, 0.145)$ for 1927–1996, and $(1.007, -0.21)$ against $(1.007, -0.22)$ for 1977–1996.

Table 5: Posterior distributions for β, samples extended through 2024. Specifications follow Table 3: A and B condition on x_0, C and D use the exact likelihood and are sampled by Metropolis-Hastings. Takeaway: in the modern subsamples every specification places almost no posterior mass below zero, in sharp contrast to the finite-sample frequentist p-value of 0.15 for 1997–2024 in Table 4; the stationarity restriction, which mattered in 1977–1996, is nearly non-binding once the samples lengthen.

		1927-2024	1952-2024	1997-2024	2005-2024
A. Conditional; ρ free	Mean	0.13	0.21	1.42	1.12
	Std. Dev.	0.10	0.13	0.65	0.84
	Skewness	0.02	0.02	0.02	0.03
	Kurtosis	3.04	3.02	3.00	3.01
	Prob($\beta \leq 0$)	0.09	0.05	0.01	0.09
B. Conditional; $\lvert\rho\rvert < 1$	Mean	0.13	0.22	1.44	1.14
	Std. Dev.	0.10	0.13	0.63	0.81
	Skewness	0.05	0.11	0.13	0.16
	Kurtosis	2.99	2.93	2.87	2.86
	Prob($\beta \leq 0$)	0.09	0.04	0.01	0.08
C. Exact; $\lvert\rho\rvert < 1$	Mean	0.14	0.12	1.24	1.36
	Std. Dev.	0.09	0.10	0.46	0.41
	Skewness	-0.02	0.47	0.32	0.33
	Kurtosis	3.01	2.95	1.94	2.11
	Prob($\beta \leq 0$)	0.06	0.12	0.00	0.00
D. Exact, alt. prior	Mean	0.09	0.07	0.94	0.56
	Std. Dev.	0.10	0.11	0.44	0.50
	Skewness	0.12	0.09	0.15	0.32
	Kurtosis	2.96	2.64	2.19	1.68
	Prob($\beta \leq 0$)	0.19	0.29	0.00	0.10

Table 6: Paper-era versus updated-era comparison of the predictive regression. The updated full sample extends the original 1927–1996 paper-era window through the latest available data, while the 1997–2024 column isolates the post-paper out-of-sample period covering the dot-com episode, the 2008 financial crisis, and the modern buyback-heavy period. The bias adjustment is evaluated at the Kendall-corrected persistence parameter, matching the adjustment used in the main Stambaugh-bias exhibit. The takeaway is that persistence remains high and the gap between naive and Monte Carlo p-values is largest in the post-paper period.

Statistic	Paper era	Updated full sample	1997-2024
Start	1927-06	1927-06	1997-01
End	1996-12	2024-12	2024-12
Obs.	834	1170	335
OLS slope $\hat{\beta}$	0.2100	0.1288	1.4252
Bias adjustment	0.0705	0.0538	0.5111
Bias-adjusted slope	0.1395	0.0750	0.9141
Raw persistence $\hat{\rho}$	0.973	0.984	0.969
Bias-corrected persistence ρ used	0.978	0.988	0.981
Naive p-value	0.063	0.092	0.015
Monte Carlo true p-value	0.177	0.237	0.149

4 Extension to 2024

Tables 4 and 5 repeat every exhibit on the extended samples. Three results stand out. First, the bias behaves as theory predicts: in the full sample T rises from 834 to 1,170 months and the simulated bias falls from 0.074 to 0.057, tracking the $1/T$ rate in equation (2). Table 6 condenses the comparison into a single view, reporting the OLS and bias-adjusted slopes, both persistence estimates, and both p-values side by side for the paper era, the full updated sample, and the post-paper period.

Second, modern slopes are not comparable to the paper's in raw units. The 1997–2024 slope of 1.43 is roughly seven times the paper's full-sample 0.21, but the variance of the predictor's innovations falls over the same period from 0.112×10^{-4} to 0.010×10^{-4}. As firms substituted repurchases for dividends the dividend yield declined to roughly 1.5% and ceased to vary. Since a slope measures return per unit of yield, a threefold contraction in the predictor's standard deviation mechanically inflates the coefficient. A standardized slope, $\hat{\beta} \times \mathrm{sd}(x)$, would be the appropriate cross-era comparison.

Third, the paper's warning binds more tightly today than in 1999. In 1997–2024 the naive p-value is 0.015—conventionally significant—while the finite-sample p-value is 0.149. The gap is tenfold, against roughly threefold in the paper's own full sample, and the simulated bias of 0.567 is about 40% of the estimated slope. The Bayesian posterior for the same window places 1% or less of its mass below zero depending on the specification, so the divergence between frameworks has widened as well. We also note that the stationarity restriction has become nearly irrelevant: the unrestricted sampler retains about 99% of draws in every modern window, against 80% in 1977–1996, because longer samples pin down ρ without help from the prior.

Table 7: Robustness comparison between the CRSP total-market value-weighted index and a NYSE-only value-weighted index rebuilt from the CRSP monthly stock file. The high correlation shows that the dividend-yield series are very similar, so the index choice has little effect on the main result.

Statistic	Value
First overlapping month	1927-06
Last overlapping month	2024-12
Correlation of D/P series	0.9940
OLS slope, total-market index	0.128833
OLS slope, NYSE stock-file index	0.143496

5 Successes and challenges

What replicated well. The parameter estimates, the simulated finite-sample moments, all sixteen Bayesian posterior cells across four specifications, and the geometry of Figure 1 reproduce within a few hundredths, including the paper's most extreme individual result.

Return definition. Our initial slopes ran roughly 50% high in the pre-war subsamples. The cause was that we had used simple excess returns while the paper's table note specifies continuously compounded returns; the difference between the two is of the order of half the return variance, which is substantial in the volatile 1927–1951 window and negligible later. Switching to log returns moved the full-sample slope from 0.31 to 0.2100 against the paper's 0.21 and left ρ and Σ essentially unchanged, confirming the diagnosis.

Vendor schema drift. The course template's CRSP index puller targets `crsp_a_indexes.msix`, the capitalization-decile file. The market index we require resides in `crsp.msi` on our WRDS instance and uses `date` rather than `caldt` as its date column. We resolved this by enumerating libraries and describing tables through the `wrds` API before writing the query rather than trusting documentation.

Sensitivity in the shortest sample. Our simulated bias for 1977–1996 is 0.52 against the paper's 0.42. At $\hat{\rho} = 0.990$ with 239 observations the bias in equation (2) is extremely sensitive to ρ, so a difference of 0.003 in the persistence estimate propagates materially. We also condition the simulation on the first observed predictor value, whereas the paper conditions on both sample endpoints.

A method that failed, and why. Our first approach to Parts C and D avoided building a sampler altogether: since the exact posterior differs from the conditional one by a single observation's contribution, we reweighted the Part B draws by the stationary density of x_0, using importance sampling. In the long, lower-persistence samples this worked well, with effective sample sizes above 90% of nominal draws. In the post-war samples it failed. Part D's prior carries a $(1 - \rho^2)^{-1}$ factor that grows without bound as ρ approaches one, so in the 1952–1996 sample the ten heaviest draws all had $\rho > 0.999$ and the weighted posterior mean of ρ was dragged from 0.981 to 0.989, displacing that specification's position in Figure 2. The diagnosis is that the conditional posterior, concentrated near $\rho = 0.98$, is a poor proposal for a target that places substantial weight against the unit-root boundary; drawing more samples would not have helped. We therefore replaced the reweighting with a Metropolis-Hastings sampler that targets the exact posterior directly, which recovers sensible persistence estimates (0.973 for Part C and 0.975 for Part D in the full sample) and matches the paper across all sixteen cells. The importance-sampling code remains in the repository, documented as the attempt that motivated the switch.

Testing without credentials. Our replication tests require the tidy panel, which requires WRDS credentials that must never enter the repository. The suite is therefore split: simulation-based tests of the bias mechanism run in any environment, including continuous integration, while data-dependent tests skip cleanly with an explanatory message when the panel is absent and run locally once `doit` has built it.

6 Conclusion

The dividend–price ratio's apparent ability to predict returns is substantially an artifact of estimation. In the paper's own sample the honest p-value is three times the naive one; in the quarter-century since, it is ten times. Whether any predictability survives depends on the inferential framework: for 1927–1996 the finite-sample frequentist test cannot reject a zero slope at 0.18 while the Bayesian posterior places between 4% and 10% of its mass below zero depending on the specification. That defensible frameworks disagree on identical data is, we think, the paper's most durable lesson.

Reproducibility

The project rebuilds end to end with `doit`, which pulls from WRDS, constructs the tidy panel, regenerates every table and figure in this document, executes the walkthrough notebook, and runs the test suite. Every statistic reported here is generated by code; none is entered by hand.